Independent review of BSGS runbook coverage

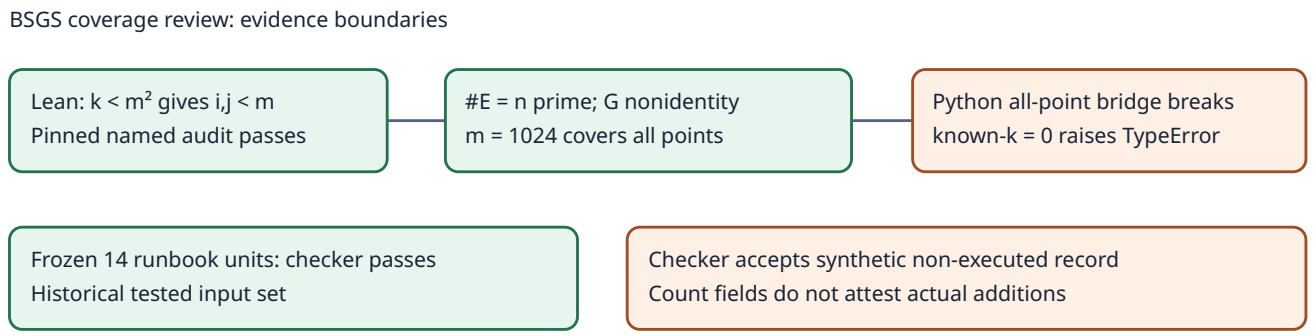
Formal and source review packet

3 November 2026

Scoped result

The pinned Lean theorem correctly decomposes every natural exponent $k < m^2$ for $m > 0$ into indices $i, j < m$. The named hosted Lean 4.35.0 build and axiom audit passed. Independent point enumeration confirms that the supplied toy curve has prime order 1,047,689 and its supplied nonidentity point generates the curve. With $m = 1024$, the mathematical indices cover every curve point.

The Python runbook's broader all-point bridge fails at the identity: `known-k=0` raises a `TypeError` before writing a result. The checker accepts a synthetic record without solver execution and an arbitrary `setup_ops`. These defects do not change the 14 frozen trial outcomes, all of which passed their pinned checks.



Independent joints

Joint	Verdict	Reproducible evidence
J2	Holds	Exact declaration, 11 input hashes, 89 workspace inputs, 10 verifier inputs, 10 pins; receipt from GitHub run 41698891260; false-boundary controls.
J3	Holds	Blind quotient/remainder derivation, exhaustive character-sum point count; width 1023 misses 1160 points.
J1	Breaks bridge	Independent curve count and 14 trial checks hold; zero-scalar crash and synthetic-record acceptance disprove the all-point execution and count bridge.

Next technical repair

Encode the identity consistently in the driver and checker and verify `known-k=0`. Separate table entries, lookup rounds, and actual group additions; attach an execution-authentic trace before using count fields as measured operations. Keep the frozen 14-unit record immutable. Full hashes, controls, first verdict attestations, and checker output are in the companion repository packet.